

File map & implementation order

10 first-run files + 2 deferred files. Work inside cuboid_translation_x/. Edit the two existing files; create the others. Reuse physics_formula.py and labels/ as dependencies.

Authority: experiment rules below follow the PDF. File boundaries, API names and stacks are implementation recommendations. Unspecified numerical choices remain open.

Order / file	Responsibility
01 config.py	Fixed protocol; unresolved run choices; seed
02 data.py	CSV loading; IDs; reference groups; scaling
03 neural_network.py [edit]	4-input / 2-output model
04 metrics.py	Full-set evaluation; rulers; groups; baselines
05 checkpoints.py	Snapshots; absolute best; reload verification
06 stopping.py	Target / progress / budget decisions
07 checks.py	Data, scaling and learning preflight checks
08 training_loop.py [edit]	Batch updates; evaluations; logging; termination
09 report.py	Curves; failures; summary; diagnosis
10 final_test.py	Frozen-choice test evaluation; separate results
11 animate.py [deferred]	First-model demonstration after result review
12 compare.py [deferred]	Controlled comparisons; separate rollout study

START CHECKLIST

Open choices: stopping ruler = coarse / intermediate / fine; absolute min_delta in standardized validation-MSE units; record a seed. No invented defaults.

Core stack: Python standard library + NumPy + PyTorch + Matplotlib. Keep experiment settings in config.py; keep generated evidence in a separate run directory.

Read downward: each numbered box is one coding task. Complete its CHECK before following the arrow. Pages 2-5 continue the same sequence.

01-03 / Data contract & model

01 OPEN config.py / PROTOCOL

Responsibility: Keep the experiment settings in one place so every file uses the same rules, numerical choices and random seed.

Stack / search: Python dataclasses; pathlib; random.seed; numpy.random.seed; torch.manual_seed.

Write: RunConfig. Inputs [v0, F, t1, t]; targets [d, v]. Train 8,000; validation 1,000; test 1,000 reserved. Fixed mass and friction. Network $4 \rightarrow 32 \rightarrow 32 \rightarrow 2$; CPU; float32.

Lock: Adam; learning rate 0.001; weight decay 0; batch 128; maximum 10,000 updates; MSE averaged over standardized outputs and samples; patience 500 updates; coverage $\geq 95\%$ on BOTH full train and validation sets. No group gate.

Rulers (d / v): coarse 0.1 m / 0.1 m/s; intermediate 0.01 m / 0.01 m/s; fine 0.001 m / 0.001 m/s. Select ONE for stopping; evaluate all three throughout.

CHECK: ruler, min_delta and seed recorded before the run. Progress clock starts at update 0; no warm-up.



02 OPEN data.py / LOAD → GROUP → SCALE

Responsibility: Turn CSV rows into checked, scaled inputs and targets for the model. Keep row IDs and motion groups attached so predictions can be traced and compared.

Stack / search: csv.DictReader; NumPy arrays; hashlib.sha256; torch.utils.data.Dataset / DataLoader.

Write: load_split(), attach_groups(), fit_scaler(), transform(), inverse_transform(). Map CSV columns to [v0, F, t1, t] and [d, v]; retain split + row ID. Check columns, finite values, valid times and shapes.

Reference groups: always resting: $v_0 = 0$ AND $\text{abs}(F) \leq \mu_s \times m \times g$; moved then stopped: not always resting AND reference final $v = 0$; moving at observation: reference final $v \neq 0$, including reversals.

Keep: exact reference rest/stop logic; disjoint, exhaustive groups; IDs and groups aligned through shuffling. Groups are metadata, never inputs or targets. Do not infer movement from net displacement. Existing CSVs need no overwrite.

Scale: fit on training rows only; handle near-zero scales; record the scale-handling choice. CHECK inverse recovery, [N,4] inputs / [N,2] targets, and rest/sliding/stopping/reversal/breakaway labels.



03 OPEN neural_network.py / PREDICT

Responsibility: Define the trainable network that maps initial conditions and time to predicted displacement and velocity, using standardized inputs and outputs.

Stack / search: PyTorch nn.Module; nn.Linear; nn.Sequential; nn.ReLU; nn.MSELoss.

Write: retain transmodel; linear widths $4 \rightarrow 32 \rightarrow 32 \rightarrow 2$. Recommendation: retain the existing hidden ReLU layers; the PDF does not specify activation. Inputs and outputs use the recorded normalization.

CHECK: CPU float32; batch output [N,2]; finite predictions. Inverse-scale outputs before physical errors or tolerance comparisons.

04-06 / Evaluation & stopping

04 OPEN metrics.py / MEASURE

Responsibility: Measure how accurate predictions are: calculate loss, physical errors and pass rates for whole datasets and motion groups, and compare simple baseline guesses.

Stack / search: torch.inference_mode; model.eval; reduction="none"; NumPy percentile; Boolean masks.

Write: evaluate(), group_metrics(), baseline_metrics(). Evaluate full train and validation with the SAME unchanged model. Standardized MSE = sum of the two squared output errors over rows / (2 × row count). Aggregate by sample counts, never average batch means.

Joint pass: $\text{abs}(d_error) < \tau_d$ AND $\text{abs}(v_error) < \tau_v$. Strict inequalities. Compute all three rates from the same predictions. Keep per-row squared errors before group aggregation.

Diagnostics: full-validation group counts, MSE, all joint pass rates; physical d/v MAE, P95 and maximum errors. Report overall physical errors too. Empty group: count 0, metrics undefined.

Extra slice: $v_0 = 0$ AND $\text{abs}(\text{abs}(F) - 3.924\text{ N}) \leq 0.2\text{ N}$; overlaps groups; no acceptance gate. At update 0 score [0,0] and $[v_0 \times t, v_0]$ baselines. Keep per-row errors at updates 0 and 10, if reached.



05 OPEN checkpoints.py / PRESERVE

Responsibility: Save and reload model snapshots and training state so results can be reproduced, the best model recovered and a run continued.

Stack / search: torch.save / torch.load; state_dict; Python json; RNG getstate / setstate.

Write: save_checkpoint(), load_checkpoint(), save_if_best(). Save every scheduled evaluation, final and selected models. Store weights, model config and normalization; optimizer and RNG states for continuation. Recommend recording update, epoch and sampler position for continuation.

Absolute best: save every new validation-MSE minimum, even an improvement smaller than min_delta. Keep this best separate from the progress reference. CHECK reload reproduces predictions.



06 OPEN stopping.py / DECIDE

Responsibility: Decide when training should stop by checking accuracy targets, lack of meaningful progress and the update limit; return the reason and model-selection decision.

Stack / search: Python dataclass; Enum; state machine. Write StopState and check_stop(); return decisions to the checkpoint writer.

State: progress-reference MSE and update; separate absolute-best MSE/checkpoint. Initialize progress reference at update 0. Inputs: update, matched overall train/validation selected-ruler rates, validation MSE.

Exact check order: finite metrics → save new absolute best → target → update progress reference → patience → cap. Numerical failure aborts separately; never success.

Target: both rates ≥95%, including update 0: TARGET_REACHED; save/select the first-passing checkpoint. Otherwise meaningful progress iff $\text{current_val_mse} < \text{reference_mse} - \text{min_delta}$; replace reference and reference update only then.

Stop: elapsed updates since reference ≥500: NO_PROGRESS. At 10,000: MAX_UPDATES. If both, record both flags; primary NO_PROGRESS. Select lowest-validation-MSE checkpoint unless target reached. Small gains accumulate against the unchanged reference; stops occur only at checks.

07-09 / Check, train & diagnose

07 OPEN checks.py / PREFLIGHT

Responsibility: Catch mistakes in data, scaling and learning before the real run, using small checks that confirm the training pipeline works.

Stack / search: Python unittest; numpy.testing.assert_allclose; torch.isfinite; tensor.clone.

Write: check_data(), check_scaling(), check_learning(). Check data and reference cases; training-only scaler and inverse recovery; one update changes weights with finite loss/gradients; substantially reduce error on 32-64 fixed examples.

CHECK: small fit is an implementation check, not generalization evidence or a perfect-boundary-fit requirement. Start fresh afterward. Recommended stop tests: target at 0; strict tolerance boundary; sub-min_delta accumulation; progress at 200 then stop at 700; patience + cap; numerical abort.



08 OPEN training_loop.py / ORCHESTRATE

Responsibility: Run the experiment: feed batches to the model, update its weights, schedule evaluation, save checkpoints, log evidence and apply stopping decisions.

Stack / search: torch.optim.Adam; DataLoader(shuffle=True, drop_last=False); time.perf_counter; csv.DictWriter.

Setup: seed → load train/validation → fit scaler → fresh model + Adam → evaluate update 0 → checkpoint + stop check. Keep test reserved.

Update: batch → predict → MSE → clear gradients → backward → optimizer update → increment global step. Shuffle each epoch; keep partial batch and already-passing samples. Record batch loss each update; abort numerical failures separately.

Evaluate: updates 0, 10, 20, 50, 100; then every 100; also normal termination. Use full matched train/validation metrics, never the latest batch pass rate. Invoke the ordered checkpoint/stop pipeline; restore train mode if continuing.

CSV: update, epoch, sample exposures, elapsed time, split, MSE and three pass rates; group metrics/counts; physical errors; breakaway slice. Retain raw observations. Count actual rows seen, including partial batches.

CHECK: first possible stagnation stop = update 500; evaluate and stop at 10,000; never run update 10,001. At exit preserve final, selected and last valid evidence; record reason and all stop flags.



09 OPEN report.py / CURVES → FAILURES → REVIEW

Responsibility: Turn saved results into plots, failure tables and a readable summary so you can see what the model learned, where it struggles and what to investigate next.

Stack / search: Matplotlib pyplot.subplots / Axes.plot; csv; NumPy percentile; json. Write plot_curves(), failure_table(), summarize_run().

Plots: validation MSE and pass rates versus updates; matched train/validation curves; full-validation group curves with counts. Pass-rate axis 0-100%. Preserve raw trends; require neither a logarithmic shape nor a plateau.

Failure tables: selected AND last valid checkpoint; row ID, input, reference group, target, prediction, physical errors, severity = max(abs(d_error)/tau_d, abs(v_error)/tau_v). Use selected ruler; keep physical units. Failed-only counts/errors; representative and worst cases. Zero failures: count 0, failed-only statistics undefined.

Run record: config, data hashes/row IDs, seed, software/device, normalization, updates/exposures/time, termination. Summary: ruler/coverage, measured rates, groups, physical errors, baselines, reason and limits.

Review: raw trend + stop settings → matched train/validation → failing groups/examples → one explanation + one controlled follow-up. Use validation; no automatic model enlargement or longer training. Timing is supporting context.

10-12 / Freeze, demonstrate & compare

10 OPEN final_test.py / FROZEN CHOICE ONLY

Responsibility: Measure the frozen, selected model on reserved test data and save separate results to assess performance on examples unused in model selection.

Stack / search: reuse data.py, metrics.py and checkpoints.py; Python json; torch.inference_mode.

Write: evaluate_frozen_choice(). Freeze the training protocol and model selection BEFORE opening the reserved 1,000 test rows for evaluation. Reuse training-fitted normalization and the selected checkpoint.

Report: separate test metrics; no tuning against test results. Training target attainment is distinct from final test acceptance; the PDF supplies no separate numerical test-acceptance threshold.

CHECK: label the scope as synthetic one-dimensional reference agreement, not general 3D or real-world validation. If validation follow-ups are still changing the choice, defer this test execution.



11 OPEN animate.py / SECOND MILESTONE

Responsibility: Show the first trained model visually alongside the analytical motion reference, using saved checkpoints after the training result has been reviewed.

Stack / search recommendation: Matplotlib animation.FuncAnimation; saved PyTorch state_dict; analytical reference.

Gate: review the first training result before animation design. Build the first-model demonstration before other experiments; Models B/C are not prerequisites.

Write later: load saved training checkpoints + recorded normalization; obtain corresponding analytical-reference predictions; implement the reviewed demonstration design.

CHECK: use the first model and its saved training progress. Animation design remains deferred as required by the PDF.



12 OPEN compare.py / AFTER THE DEMONSTRATION

Responsibility: Run fair follow-up experiments that change one main factor at a time, such as network size or training-data size, and compare the measured results.

Stack / search: Python dataclasses.replace; NumPy subset indexing; reuse training/evaluation/report modules; Matplotlib.

Write later: run_architecture_comparison(), run_data_comparison(), run_loss_comparison(), study_rollouts(). Change one main factor at a time; preserve matched data and protocols.

Architectures: 4 → 16 → 2; 4 → 32 → 2; 4 → 16 → 16 → 2; 4 → 32 → 32 → 2. Repeat promising comparisons with three seeds.

Data: nested 500 / 1,000 / 2,000 / 4,000 / 8,000 training rows; fresh models; subset-specific normalization. Loss comparisons are deferred; candidate losses are not specified by this PDF.

Rollouts: separate study; matched force schedules; explicit position tracking; out-of-range checks. The PDF refers to supporting notes for comparison limitations; do not invent additional comparison standards in this manual.

CHECK: first-run training + evidence → result review → first-model animation → controlled comparisons. Keep final test evaluation tied to a frozen choice, not a recurring tuning loop.

Coding manual amendment

27 September 2026 | PyTorch scheduling and checkpoints

Confirmed 2026-09-27. This amendment supersedes the earlier constant-learning-rate assumption in the plan and coding manual. Training-loop integration remains pending.

Two patience rules

Use PyTorch `torch.optim.lr_scheduler.ReduceLROnPlateau` with Adam. Initial learning rate: 0.001; factor: 0.5; minimum: 0.00001; cooldown: one scheduler check. Monitor full-validation standardized MSE with `mode="min"`, `threshold_mode="abs"`, and `threshold=config.min_delta` (currently 0.001).

Learning-rate reduction patience is 200 optimizer updates. Early-stopping patience remains 500 optimizer updates. Call the scheduler at update 0 to establish its baseline, then at updates 100, 200, 300, etc. With this regular cadence, set PyTorch `patience=1`: it reduces on the second unsuccessful check, after 200 updates. `patience=2` would wait 300 updates, and `patience=200` would count checks rather than optimizer updates. Early evaluations at 10, 20 and 50 are still saved and used by the stop checker, but do not advance the scheduler.

If both progress references were last improved at update 1000, a continuous stall reduces the rate at 1200 and can stop at 1500. This leaves 300 optimizer updates after the first reduction. Cooldown and subsequent reductions have their own scheduler counters. Early irregular evaluations can move the stopping reference independently, so this aligned example is not a universal guarantee. Keep both references in the logs.

The scheduler and early stopper compete for the same training budget: stopping before or at a reduction gives the new rate no opportunity. A reduction alone never resets the 500-update stopping clock. A meaningful validation-MSE improvement does. Preserve the 10000-update cap and the first-target success rule. At a terminal check, do not schedule a reduction for training that will not run. Check finite metrics and rank snapshots; decide target/progress/patience/cap; on continuation, step the scheduler at its cadence; persist the resulting optimizer, scheduler and stopping state before the next batch update.

Reducing learning rate is an optimization attempt, not a diagnosis or cure for overfitting. An absolute threshold of 0.001 becomes relatively demanding as MSE shrinks; inspect real curves before revising it. Save every new absolute MSE minimum even if its gain is smaller than this threshold.

Checkpoint responsibility and integration

`checkpoints.py` saves immutable `training_data/<run>/checkpoints/update_XXXXXX.pt` files with weights, model/run configuration, normalization, validation and optional training metrics, data identity, optional Adam and scheduler states, CPU random states, and caller-provided continuation metadata. `index.json` stores `best_mse`, `best_pass_rate`, `last`, and explicitly selected pointers. Highest pass rate uses the selected ruler and lower MSE as its tie-break; exact ties retain the earlier state. The two best criteria may identify different models. The manual's first-target/lowest-MSE fallback remains the final-selection policy until explicitly revised.

For exact CPU continuation, the future training loop must supply and restore epoch, current shuffled permutation, next batch position, any dedicated data-generator state, exposure count, and stopping state. Loading weights alone restores prediction behavior, not the training trajectory. Scheduler control belongs to the training loop; checkpoint saving does not autonomously train, reduce the rate or roll back. Rollback-and-retry requires a new run linked to its parent snapshot.

Record the learning rate used by updates, rate changes, and the next rate after each evaluation. Save scheduler state after its decision. Verify prediction round trips and Adam/scheduler continuation before training. The test split never participates in ranking or scheduling.

Source: Official PyTorch `ReduceLROnPlateau` API documentation.